

Project 15

Lifting a custom__3R Posture-Change Loop to custom__6R

Format: Group of two students; simulation-based project.

Course foundations: Lectures 3–7; Exercises 2–3.

Motivation

A positioning arm may execute a nonsingular posture change while its attached wrist cannot maintain the desired tool orientation without becoming singular or reaching a limit. Studying the complete robot reveals the gap between a feasible wrist-center loop and a feasible full-pose motion.

Task

Overall objectives. Extend a supplied nonsingular custom__3R arm connection to the full custom__6R robot and determine how wrist IK, orientation, and limits affect the resulting trajectory.

- **Match the arm and full-robot models.** Verify that the custom__6R positioning subchain has the custom__3R geometry and identify the wrist center. Document any fixed transform between the custom__3R operational point and that center. Validate the D–H/URDF models before transferring a trajectory.
- **Inspect the supplied arm loop.** Verify that the supplied joint path connects distinct arm IKs of the same wrist-center position and remains regular in sampled checks. Plot $\det J_{A,v}$ in (q_2, q_3) , overlay the path and endpoint IKs, and map the corresponding two-/four-arm-IK workspace regions.
- **Build the full-pose command.** Choose a fixed desired tool orientation and derive tool position from the wrist-center path and the actual fixed tool transform. Repeat with one alternative fixed orientation. Do not confuse a wrist-center loop with a tool-tip position command.
- **Recover continuous wrist solutions.** For each arm sample, solve the residual wrist rotation and continue both admissible starting wrist flips. Track angle periodicity consistently. Monitor pose residuals and the arm and wrist factors in $\det J = \det J_{A,v} \det J_{W,\omega}$.
- **Compare arm and full-pose feasibility.** Identify whether each lift completes, reaches a wrist singularity, or violates a joint limit. Compare endpoint full poses and joint configurations, and refine any critical interval. Explain why one unsuccessful fixed-orientation lift is not a proof that the full robot lacks a nonsingular connection.

Specifics and scope

Use one supplied arm loop, two fixed tool orientations, and both starting wrist flips where regular. The arm loop and arm IK will be provided, so the new work is the wrist completion and validation. Report mathematical regularity and limit-admissible feasibility separately. No general spatial path optimizer or search for a new cuspidal geometry is required.

Check both translation and orientation residuals for every claimed full-pose IK. Document angle periodicity and limits, Jacobian conventions, and any scaling used for singular values. State all fixed coordinates in reduced plots; a two-dimensional slice does not show the entire 6R singular set. Refine decisive transitions, and distinguish sampled evidence from a continuous-path guarantee. Collision freedom is not implied by these kinematic checks.

Expected outcome

Understand the custom_3R-plus-wrist construction and explain how positioning-arm topology interacts with wrist orientation. Deliver four candidate-lift records where applicable, paired arm maps, full-pose closure checks, and a clear comparison of arm versus full-robot failure mechanisms.

Submit reproducible code, model parameters and test data, the required plots, a concise 5–6-page report, and a short simulation demonstration. Explain both successful cases and failures; conclusions must follow the numerical and geometric evidence rather than an expected outcome.

Resources

The custom_6R consists of the custom_3R positioning arm plus a spherical wrist. The arm is orthogonal with non-intersecting consecutive axes. Its URDF is `lectures_main/assets/models/custom_6R/custom_6R_new.urdf`; the corresponding STL meshes are in that directory. Inspect wrist-axis concurrency and base/tool transforms using the lecture website.

The exact custom_3R is an orthogonal 3R robot with non-intersecting consecutive joint axes. Its URDF is `lectures_main/assets/models/custom_3R/custom_3R_new.urdf`; STL meshes are in the same directory. Use the lecture website to inspect axes and frames, verify D–H parameters, plot IKs, and animate configurations. The course IK methods in Lectures 3–4 and the exact-model connection in Lecture 6 provide the starting point. A validated arm-loop seed will be supplied; adapting an idealized-atlas loop without revalidation is insufficient.

Use the lecture website to verify D–H parameters, visualize IKs and singularities, and simulate paths. All listed paths are relative to the repository root. Start the website following `lectures_main/README.md`.

Workload and collaboration

Budget approximately 60 hours per student (120 person-hours per pair) for this 2-ECTS project, following EPFL’s 30-hours-per-credit convention.¹ Deduct any lectures or exercises included in those credits. Allocate roughly 20 team-hours to model validation, 50 to the core work, 30 to experiments, and 20 to reporting. Reuse the specified IK and simulations. Share validation and interpretation even if modelling and implementation are divided between students. Hardware execution is outside scope.

¹EPFL: Design a Course, workload guidance.